Synthetic Homotopy Theory

Exercises and Solutions for the

Midlands Graduate School (MGS)
13–17 April 2026, Nottingham, UK

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Heavily based on notes from MGS 2024 by Ulrik Buchholtz and Mark Williams

NB Subject to change: Refresh if needed

Contents

1 Truncated and connected types **1**
1.1 Exercises 1

2 A menagerie of higher inductive types **4**
2.1 Exercises 4

3 The Hopf fibration and the LES **7**
3.1 Exercises 7

Chapter 1

Truncated and connected types

1.1 Exercises

1. Show that for $n \geq m$ and any type A we have

$$\|\|A\|_n\|_m \simeq \|A\|_m.$$

2. Show that a type X is n-connected if and only if, for every n-type Y, the diagonal map

$$\delta : Y \to (X \to Y), \quad y \mapsto_\delta (x \mapsto y),$$

is an equivalence.

3. Show that the map $|_|_n : X \to \|X\|_n$ is n-connected.

4. Show that a type X is $(n+1)$-type if and only if the diagonal map of X,

$$\delta_X : X \to X \times X, \quad x \mapsto (x, x),$$

is n-truncated.

5. Consider a map $f : X \to Y$. We define its **n-image** to be the type $\mathrm{im}_n(f) := (y : Y) \times \|\mathrm{fib}_f(y)\|_n$.

   a) Show that we can factorise f through the n-image, that is, construct a commuting diagram:

$$\begin{tikzcd} X \arrow[rr, "f"] \arrow[dr, "p"'] & & Y \\ & \mathrm{im}_n(f) \arrow[ur, "i"'] & \end{tikzcd}$$

   b) Show that p is n-connected and i is n-truncated.

c) Show that if Z is any type, $g : X \to Z$ is n-connected, and $h : Z \to Y$ is n-truncated with $f = h \circ g$, then there is a unique equivalence $e : \mathrm{im}_n(f) \to Z$ with $e \circ p = g$ and $h \circ e = i$:

$$\begin{tikzcd}
 & Z \arrow[dr, "h"] \arrow[dd, dotted, "e"] & \\
X \arrow[ur, "g"] \arrow[rr, "f"] \arrow[dr, "p"'] & & Y \\
 & \mathrm{im}_n(f) \arrow[ur, "i"'] &
\end{tikzcd}$$

(*Hint*) It's easier if you assume g and h are projections of dependent sums.

6. (Eckmann–Hilton). All paths will take place in a fixed type X.

a) Given paths $p_0, p_1 : x = y$ and $q_0, q_1 : y = z$, and higher paths $r : p_0 = p_1$ and $s : q_0 = q_1$, define their **horizontal multiplication** $s * r : q_0 \cdot p_0 = q_1 \cdot p_1$ by path induction on r and p_0.

$$\begin{tikzcd}
x \arrow[r, bend left, "p_0"] \arrow[r, bend right, "p_1"'] \arrow[r, phantom, "\Downarrow r"] & y \arrow[r, bend left, "q_0"] \arrow[r, bend right, "q_1"'] \arrow[r, phantom, "\Downarrow s"] & z
\end{tikzcd}$$

b) Given paths fitting into the following diagram

$$\begin{tikzcd}
x \arrow[r, bend left, "p_0"] \arrow[r, bend right, "p_1"'] \arrow[r, phantom, "r_0\Downarrow"] \arrow[r, phantom, "r_1\Downarrow"] & y \arrow[r, bend left, "q_0"] \arrow[r, bend right, "q_1"'] \arrow[r, phantom, "\Downarrow s_0"] \arrow[r, phantom, "\Downarrow s_1"] & z
\end{tikzcd}$$

show we have the **interchange law**:

$$(s_1 * r_1) \cdot (s_0 * r_0) = (s_1 \cdot s_0) * (r_1 \cdot r_0)$$

c) Deduce that for $n \geq 2$, the path composition in $\Omega^n(X, x)$ is commutative, and thus that $\pi_n(X, x)$ is abelian.

7. (The Fundamental Theorem of Identity Types)

Let $R : X \to X \to \mathcal{U}$ be a reflexive binary relation. The following are equivalent

a) The obvious map[1] $x =_A y \to R(x, y)$ is an equivalence.

[1] i.e. the map defined by path induction which sends refl_x to the term of type $R(x, x)$ given to us by reflexivity of R

b) For any $x : X$, the total space $(y : X) \times (R(x, y))$ is contractible.

8. Recall that $\mathcal{U}^{\leq n} := (X : \mathcal{U}) \times \text{is-}n\text{-type}\ (X)$ is an $(n+1)$-type. This allows us to define $R : \|X\|_{n+1} \times \|X\|_{n+1} \to \mathcal{U}^{\leq n}$ by the truncation elimination:

$$R(|x|_{n+1}, |y|_{n+1}) := \|x = y\|_n$$

Apply the fundamental theorem of identity types with the above family to deduce that $(|x|_{n+1} =_{\|X\|_{n+1}} |y|_{n+1}) \simeq \|x = y\|_n$

9. Show that when X is connected, that $\pi_n(X, x_0) \cong \pi_n(X, x_1)$ are merely isomorphic for any $x_0, x_1 : X$, that is, construct an element of the type

$$(x_0, x_1 : X) \to \|\pi_n(X, x_0) \simeq \pi_n(X, x_1)\|.$$

10. We define the observational equality type family of the natural numbers $E : \mathbb{N} \to \mathbb{N} \to \mathcal{U}$ by

$$\begin{aligned} E(0, 0) &= 1 \\ E(\text{succ}(n), 0) &= 0 \\ E(0, \text{succ}(n)) &= 0 \\ E(\text{succ}(n), \text{succ}(m)) &= E(n, m) \end{aligned}$$

a) Define a natural map $n = m \to E(n, m)$ for all $n, m : \mathbb{N}$.

b) Use the fundamental theorem of identity types to show this natural map is an equivalence.

11. Show that $\|X\|_0$ satisfies the universal property of the set quotient of X modulo the equivalence relation $\sim$, where $x \sim x'$ if and only if $\|x =_X x'\|$.

Chapter 2

A menagerie of higher inductive types

2.1 Exercises

1. (Pushout-Pullback Duality) Consider a commutating square:

$$\begin{tikzcd} A \arrow[r, "g"] \arrow[d, "f"'] & C \arrow[d, "j"] \\ B \arrow[r, "i"'] & X \end{tikzcd}$$

   with $H : i \circ f \sim j \circ g$. Then the following are equivalent:

   (i) The square is a pushout square.

   (ii) the square

$$\begin{tikzcd} T^X \arrow[r, "-\circ j"] \arrow[d, "-\circ i"'] & T^C \arrow[d, "-\circ g"] \\ T^B \arrow[r, "-\circ f"'] & T^A \end{tikzcd}$$

   is a pullback square, for every type T.

2. Show that the two definitions of S^1 are equivalent, in two ways:

   (i) Show the HIT for S^1 is the same thing as $\Sigma 2$ by constructing an equivalence between them.

   (ii) By applying Exercise 2.1.

3. Show that evaluation at the loop gives pointed equivalence

$$(S^1 \to_\star X) \to_\star \Omega X$$

for any pointed type X.

Conclude, using the suspension–loop adjunction, that we have a natural equivalence

$$(S^n \to_\star X) \to_\star \Omega^n X$$

for all $n : \mathbb{N}$.

4. Show that a type X is a proposition iff $\mathrm{inl} : X \to X * X$ is an equivalence.

5. Show that $A \vee B \simeq A * B$ for propositions A, B, where $A \vee B := \|A + B\|$. (Beware the notational overload between the wedge sum and disjunction: It's not too bad in practice, because the only pointed proposition is the unit type!)

6. (i) Consider a span $A \xleftarrow{f} X \xrightarrow{g} B$. Show that the following square is pushout:

$$\begin{tikzcd} X + X \arrow[r, "{[\mathrm{id},\mathrm{id}]}"] \arrow[d, "f+g"'] \arrow[dr, phantom, "\ulcorner"] & X \arrow[d, "inr \circ g"] \\ A + B \arrow[r, "{[\mathrm{inl},\mathrm{inr}]}"'] & A \amalg_X B \end{tikzcd}$$

   (ii) Apply this to show that for every type X we get $\Sigma X \simeq 2 * X$.

7. (Smash Product - Pointed Map Space Adjunction)

   (i) Prove that for each pointed type B, the fucnction $(-) \wedge B$ is the left adjoint to the function $B \to_* (-)$,i.e. for every pointed types A and C, there is a pointed equivalence:

$$A \to_* (B \to_* C) \simeq_* (A \wedge B) \to_* C$$

   (ii) Use this to prove that $S^1 \wedge X \simeq \Sigma X$ for each type X.

8. Show that the smash product is equivalently the pushout:

$$\begin{tikzcd} X + Y \arrow[r] \arrow[d] & X \times Y \arrow[d] \\ 1 + 1 \arrow[r] & X \wedge Y \end{tikzcd}$$

9. (i) For all pointed types X, Y, and A, prove there is a pointed equivalence:

$$X * Y \to_* A \simeq_* X \to (Y \to_* \Omega A)$$

   (ii) For pointed types X, Y prove that

$$X * Y \simeq_* \Sigma(X \wedge Y).$$

10. We can define the unit interval type $\mathbb{I}$ to be the pushout of the constant span $1 \leftarrow 1 \rightarrow 1$. Working in Martin-Löf type theory *without* univalence (and function extensionality) but with pushouts with strict definitional computation rules, prove that

    (i) $[((x, y) : A \times A) \times (x = y)] \simeq (\mathbb{I} \rightarrow A)$

    (ii) Use the above equivalence (or, if you prefer, a variation thereof) to derive function extensionality.

11. (i) Show that if the square

$$\begin{tikzcd} A \arrow[r, "g"] \arrow[d, "f"'] & C \arrow[d, "j"] \\ B \arrow[r, "i"'] & X \end{tikzcd}$$

    is a pullback, then so is

$$\begin{tikzcd} \Omega A \arrow[r, "\Omega g"] \arrow[d, "\Omega f"'] & \Omega C \arrow[d, "\Omega j"] \\ \Omega B \arrow[r, "\Omega i"'] & \Omega X \end{tikzcd}$$

    (ii) For a map $A \xrightarrow{f} B$ show that $\Omega\,(\mathrm{fib}\ f) \simeq \mathrm{fib}\,(\Omega f)$

12. Show that the square

$$\begin{tikzcd} S^1 \arrow[r] \arrow[d] & 1 \arrow[d] \\ S^1 \times S^1 \arrow[r] & S^2 \vee S^1 \end{tikzcd}$$

    is a pushout.

Chapter 3

The Hopf fibration and the long exact sequence

3.1 Exercises

1. Construct the multiplication on the circle $\mu : S^1 \to (S^1 \to S^1)$ by circle induction. Show that for each $z : S^1$ the map $\mu(z, -) : S^1 \to S^1$ is an equivalence.

2. Construct a retraction $\rho : \Omega S^2 \to S^1$ of $\eta : S^1 \to \Omega S^2$. (*Hint*) Use the Hopf fibration.

3. (i) Show that if the square

$$\begin{tikzcd} A \arrow[r, "g"] \arrow[d, "f"'] & C \arrow[d, "j"] \\ B \arrow[r, "i"'] & X \end{tikzcd}$$

is a pullback, then so is

$$\begin{tikzcd} \Omega A \arrow[r, "\Omega g"] \arrow[d, "\Omega f"'] & \Omega C \arrow[d, "\Omega j"] \\ \Omega B \arrow[r, "\Omega i"'] & \Omega X \end{tikzcd}$$

(ii) For a map $A \xrightarrow{f} B$ show that $\Omega\,(\mathrm{fib}\ f) \simeq \mathrm{fib}\,(\Omega f)$

(iii) Show that if $A \xrightarrow{f} B \xrightarrow{g} X$ is a fiber sequence, then $\Omega A \xrightarrow{\Omega f} \Omega B \xrightarrow{\Omega g} \Omega X$ is a fiber sequence.

4. (Flattening Lemma) Given a span $A \xleftarrow{f} C \xrightarrow{g} B$ with pushout D and a type family $P : D \to \mathcal{U}$ defined by

$$\begin{aligned} P_{\text{inl}} &: A \to \mathcal{U} \\ P_{\text{inr}} &: B \to \mathcal{U} \\ P_{\text{glue}} &: (c : C) \to P_{\text{inl}}(f(c)) \simeq P_{\text{inr}}(g(c)), \end{aligned}$$

the total space of P is equivalent to the pushout of the span

$$\big((a : A) \times P_{\text{inl}}(a)\big) \longleftarrow \big((c : C) \times P_{\text{inl}}(f(c))\big) \longrightarrow \big((b : B) \times P_{\text{inr}}(b)\big),$$

where the left maps is $(c, u) \mapsto (f(c), u)$ and the right map is $(c, u) \mapsto (g(c), P_{\text{glue}}(c)(u))$.

5. (i) Find a pointed type X and a map $S^2 \xrightarrow{f} X$ such that the fiber of f is S^3.

   (ii) Is there any pointed map out of S^1 such that its fiber is S^2?